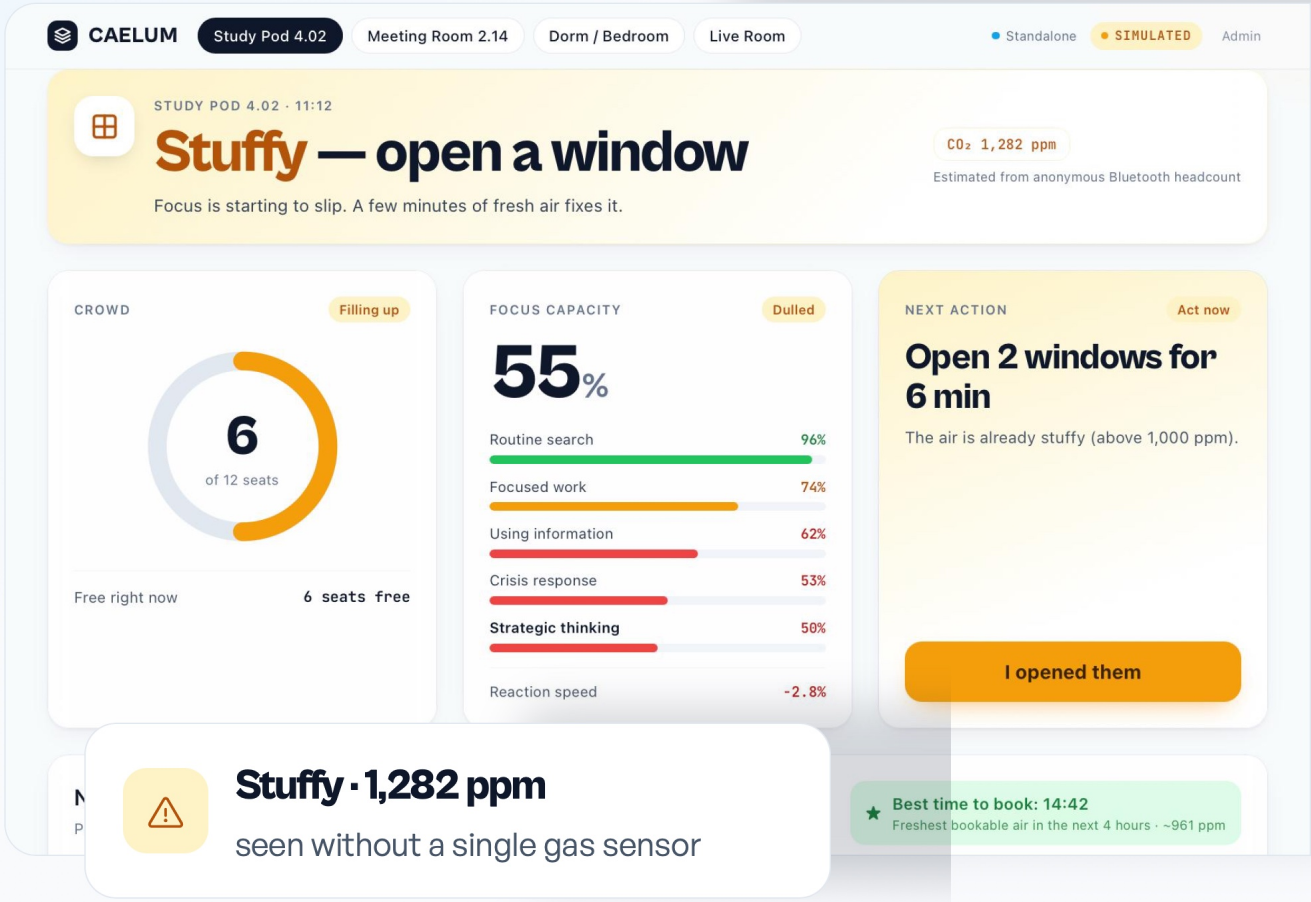
VIRTUAL CO_2 SENSOR · LIVE DEMO

CO_2 is invisible. Its cost isn't.

Caelum turns anonymous Bluetooth signals into a live virtual CO_2 sensor, showing the gas you can't see and the focus it quietly takes away.

- Hidden signal
- Hidden gas
- Hidden cost



What the invisible gas costs you

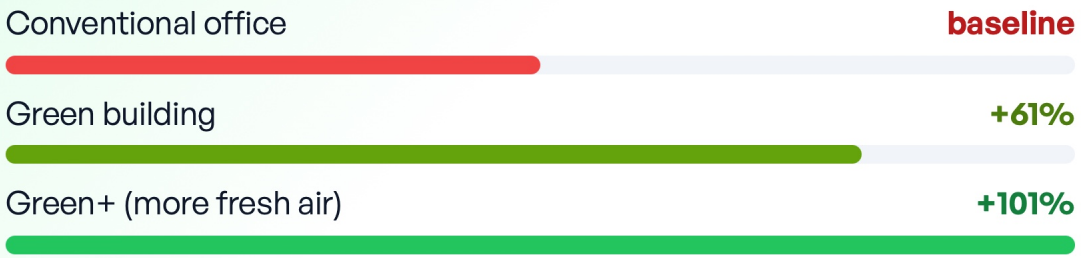
What controlled studies measured about a gas nobody can see.

COGNITIVE FUNCTION

Harvard COGfx 2016

+101%

higher cognitive scores with more fresh air



Odourless

0 senses

notice CO_2. It just feels like brain fog.

Allen 2016

−21%

cognitive score for every extra 400 ppm of CO_2

LBNL 2012

1,000 ppm

is where decision-making starts to slip

MacNaughton 2015

$6,500 / worker

a year in productivity, for under $40 of air

VIRTUAL SENSORS

Nothing to mount

Cheaper wherever an estimate is enough

One reusable Bluetooth scanner per area replaces a wired CO_2 sensor in every room: no install, no calibration, no removal.

DTU 2016

Better sleep

Lower bedroom CO_2, sharper next-day thinking

Private

No cameras

Anonymous Bluetooth counts. No faces, no names.

How we make CO_2 visible

Three hidden layers, one screen you read in three seconds.

01 Hidden signal
Bluetooth packets reveal who is in the room

→

02 Hidden gas
Mass-balance physics turns people into ppm

→

03 Hidden cost
The Harvard curve turns ppm into lost focus

→

04 Made visible
A colour, a number and one thing to do

Live

Bluetooth headcount

Counts phones in range, skips earbuds and trackers.
No cameras.

Physics

Virtual CO_2 sensor

Live ppm from people, room volume and windows. No gas sensor.

COGfx

Focus capacity

What the air costs: focus % per task, from routine to strategy.

Room view

3-second glance

Fresh or stuffy in plain words and colour, readable from the door.

4 boards

Room heatmap

Four corner scanners show where people sit, not just how many.

Model

4-hour forecast

See the stuffy hour coming, with an uncertainty band around it.

Act now

Next action

“Open 2 windows for 6 min”, plus fresh-air breaks in the calendar.

Admin

Ops console

Live packet log, scanner heartbeats and range tuning.